Applied Statistics: Time Series

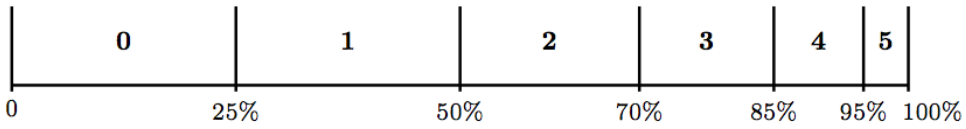
Mock exam

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Name: Surname:

In this exercise, you are asked to estimate your degree of certainty. Please use the following scale:



Each question will be graded according to the following scale:

		Degree of certainty					
		0	1	2	3	4	5
Answer	Correct	+13	+16	+17	+18	+19	+20
	Incorrect	+4	+3	+2	0	-6	-20

The exercise score will be calculated as follows:

$$\text{Score } = \max\left\{0, \frac{\text{Total points}}{10}\right\}.$$

Provide your answers along with the respective degree of certainty (without justification), in the followin table:

Question	1	2	3	4	5	6	7	8	9	10
Answer										
Degree of certainty										

Tournez la page.

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Question 1

When setting up a monitoring device, how do you choose the sampling rate?

A. Sample as fast as the hardware permits, since a finer time step is always an improvement and allows for better analysis.

B. Make the time step as small as needed for the phenomenon of interest, but as large as possible to spare memory and computation.

C. The time step is imposed by the sensor manufacturer and cannot be chosen.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Question 2

Consider the time series X,

Time (s)	0	1	2	3	4	5
X	0.2	0.3	0.1	0.1	0.5	0.3

Which one(s) of the following time series is a resampling of X ?

A.

Time (s)	0	2	4
Y	0.2	0.1	0.5

B.

Time (s)	0	2	4
Y	0.25	0.1	0.4

C.

Time (s)	0	2	4
Y	0.3	0.1	0.3

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Tournez la page.

Question 3

What is the strength of the mean compared to the median?

A. Discards outliers.

B. More robust to missing data.

C. Easy to compute.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Question 4

A colleague computes covariances between several pairs of time series from an experiment (temperature, pH, oxygen concentration, aeration rate) and finds the numbers impossible to compare with each other.

Why does switching to correlation resolve this?

A. Correlation detects non-linear relationships that covariance misses.

B. Correlation ignores the sign of the relationship, which covariance does not.

C. Correlation is computed on fewer data points, so it is less noisy.

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.

Question 5

When computing a moving average, what should you keep in mind?

A The time series get longer after processing.

B The general trend of the time series disappears in the process.

C The time series' average value is almost unchanged.

D None of the above answers.

E The three answers A, B, and C are correct.

F The information in the question is not sufficient to answer.

Question 6

Using a portable device, you have measured the heart rate of a patient every 10 minutes over a period of 60 days. This gives you a time series $X = (X_0, X_1, X_2, ..., X_{8639})$, where the value X_t is the patient's average heart rate between time t and $t + 5$ minutes.

Computing the ACF (Autocorrelation function) of X, what do you expect to observe?

A. A peak every minute, corresponding to the reset of the measuring device.

B. A peak every ten minutes, corresponding to the recording of the new measurement.

C. A peak every six hours approximately, corresponding to the physiological change of digestion (the patient have their meal at 7h, 13h, and 19h).

D. None of the above answers.

E. The three answers A, B, and C are correct.

F. The information in the question is not sufficient to answer.